SciLifeLab

Developing AI Agents in Life Sciences

A Hands-on Workshop

March 5, 2026

repo: https://shorturl.at/TLx0j

Introductions

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AGENDA

10.00-10.15 Introduction

10.15-10.30 Demo: AI agents in action

10.30-12.00 Hands-on session 1: Developing AI agents with LangChain & ReAct

12.00-13.00 Lunch

13.00-13.30 Continuation of Hands-on 1: Developing AI agents with LangChain & ReAct

13.30-15.00 Hands-on session 2: AI agent collaboration with the Model Context Protocol (MCP)

15.00-15.15 Deploying AI Agents: Legal, Technical, and Practical Considerations

15.15-15.30 Concluding Session: Open Q&A

15.30 Fika and networking

SciLifeLab

Exploring how AI agents can transform life sciences

Introduction and demos

What is an AI Agent?

AI Agent

[AI] [Agent]

AI model:

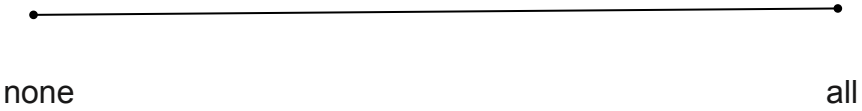
- LLM primarily
- not only LLM, also RL, etc

Agent:

- *a person who acts for or represents another*
- *a person or thing that produces a particular effect or change*
- *(also: secret agent, foreign agent)*

Agentic AI

The more an AI model (LLM) decides the behaviour of a system, the more agentic it is.



Agentic AI

What do we mean *decides?*

- Decides what information to monitor
- Decides what is relevant
- Decides what action to try (execute)
- Decides if done, or continue

Agentic AI

What sets Agentic AI apart

- May act autonomously, adapt and pursue goals
- Can be proactive, plan, self-correct
- Iterate
- Flexibility, adjusting to new environments or objectives
- Manage workflows, autonomous drones, self-improving systems
- Remembers information, preferences
- Intentional, goal-driven

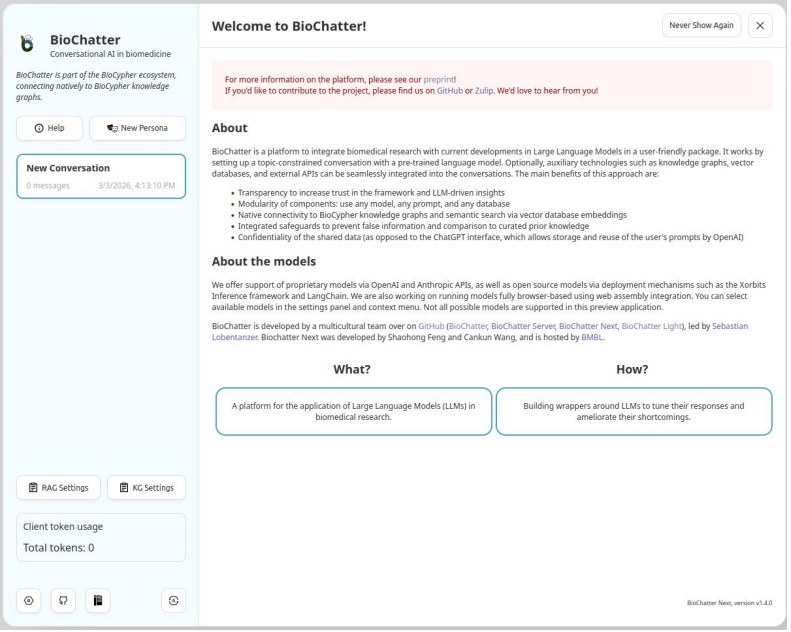
Traditional software vs AI Agent

Trait	Traditional Software	AI Agent
Context	Operates within predefined inputs/outputs.	Adapts to dynamic, real-world contexts.
Learning	Static; updates require manual coding.	Continuous; learns from data, feedback, or experience.
Initiation	Started/stopped by users or scripts.	Can self-initiate tasks based on goals or triggers.
Decision-Making	Rule-based or algorithmic (deterministic).	Autonomous, probabilistic, or goal-driven.
Flexibility	Rigid; limited to programmed functions.	Adapts behavior to new scenarios or objectives.

Real-world AI Agent Examples

BioChatter and BioCypher

General-purpose AI Agents within life science



Correspondence | Published: 22 January 2025

A platform for the biomedical application of large language models

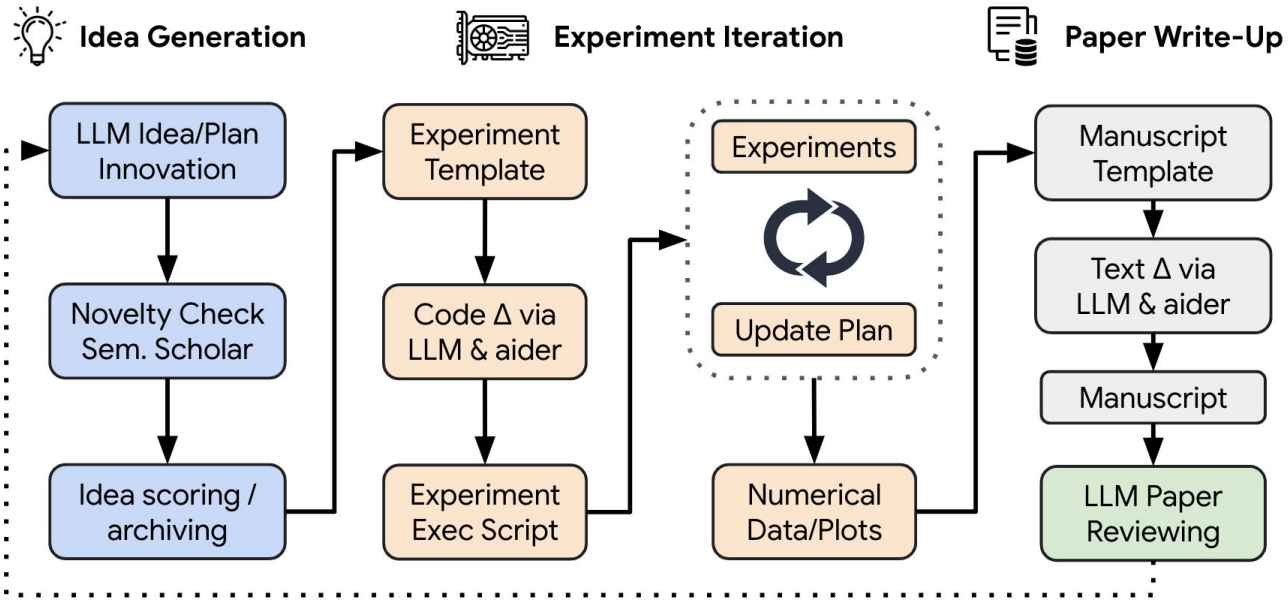
Sebastian Lobentanzer, Shaohong Feng, Noah Bruderer, Andreas Maier, The BioChatter Consortium, Cankun Wang, Jan Baumbach, Jorge Abreu-Vicente, Nils Krehl, Qin Ma, Thomas Lemberger & Julio Saez-Rodriguez

Nature Biotechnology **43**, 166–169 (2025) | Cite this article

34k Accesses | **58** Citations | **185** Altmetric | Metrics

Generative artificial intelligence (AI) has advanced considerably in recent years, particularly in the domain of language. However, despite its rapid commodification, its use in biomedical research is still in its infancy[1,2]. The two main avenues for using large language models (LLMs) are end-user-ready platforms, which are usually provided by large corporations, and custom solutions developed by individual researchers with programming knowledge. Both use cases have significant limitations. Commercial platforms do not meet the transparency standards required for reproducible research; none are open source, and only a few provide (superficial) scientific descriptions of their algorithms[3]. They are also subject to privacy concerns (reuse of user data) and to considerable commercial pressures. In addition, they are not fully customizable to accommodate a specific research domain or workflow

Sakana AI



The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

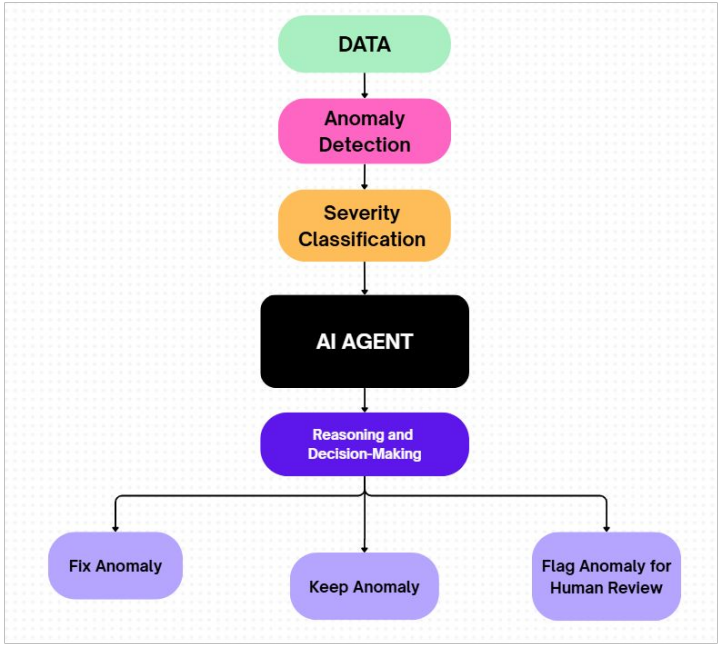
Chris Lu[1,2,*], Cong Lu[3,4,*], Robert Tjarko Lange[1,*], Jakob Foerster[2,†], Jeff Clune[3,4,5,†] and David Ha[1,†]
[1]Equal Contribution, [2]Sakana AI, [3]FLAIR, University of Oxford, [4]University of British Columbia, [5]Vector Institute, [6]Canada CIFAR AI Chair, [†]Equal Advising

One of the grand challenges of artificial general intelligence is developing agents capable of conducting scientific research and discovering new knowledge. While frontier models have already been used as aides to human scientists, e.g. for brainstorming ideas, writing code, or prediction tasks, they still conduct only a small part of the scientific process. This paper presents the first comprehensive framework for fully *automatic scientific discovery*, enabling frontier large language models (LLMs) to perform research independently and communicate their findings. We introduce THE AI SCIENTIST, which generates novel research ideas, writes code, executes experiments, visualizes results, describes its findings by writing a full scientific paper, and then runs a simulated review process for evaluation. In principle, this process can be repeated to iteratively develop ideas in an open-ended fashion and add them to a growing archive of knowledge, acting like the human scientific community. We demonstrate the versatility of this approach by applying it to three distinct subfields of machine learning: diffusion modeling, transformer-based language modeling, and learning dynamics. Each idea is implemented and developed into a full paper at a meager cost of less than $15 per paper, illustrating the potential for our framework to democratize research and significantly accelerate scientific progress. To evaluate the generated papers, we design and validate an automated reviewer, which we show achieves near-human performance in evaluating paper scores. THE AI SCIENTIST can produce papers that exceed the acceptance threshold at a top machine learning conference as judged by our automated reviewer. This approach signifies the beginning of a new era in scientific discovery in machine learning: bringing the transformative benefits of AI agents to the *entire* research process of AI itself, and taking us closer to a world where *endless affordable creativity and innovation* can be unleashed on the world's most challenging problems. Our code is open-sourced at https://github.com/SakanaAI/AI-Scientist.

06292v3 [cs.AI] 1 Sep 2024

Developing agents capable of conducting scientific research and discovering new knowledge.
Source: https://sakana.ai/ai-scientist/

A smaller, niche, real-time example



COVID-19 Anomaly Detection with an AI Agent

Near real-time anomaly detection and handling

Reads live, time-series data via API

Detects

- unusual behavior
- checks context

Then **decide** whether:

- to fix the data, keep it as a real signal, or flag it for review

Reduces manual intervention and delay

Approach could learn from human feedback

Accessed March 3, 2026.
https://towardsdatascience.com/building-an-ai-agent-to-detect-and-handle-anomalies-in-time-series-data/

AI Agent Benefits

AI agents may:

- adapt experiments based on interim results
- complement information from other domains
- connect disparate systems or tools faster as prototypes
- act on new information or context
- automate workflows

Demo time: Time to Unleash the Agents!

Created with Mistral AI